Yoonhyuk CHOI

PORTFOLIO

Contents

1. Curriculum Vitae

2. Work Experience

3. Projects

4. Future Work

Curriculum Vitae

Yoonhyuk CHOI

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Postdoc Researcher

- Arizona State University, Tempe (2023. 11. 11 – 현재)

Research Area (주요 연구 분야)

- Recommender System (추천 시스템)

  * Personalized recommendation, user-item interaction modeling

- Graph Theory (그래프 이론)

  * Social network analysis, message-passing, graph heterophily

- Large Language Model (대규모 언어 모델)

  * Text summarization, information theory

Work Experience

M.S. and Ph.D.

- 서울대학교 (2019. 03. 01 – 2023. 08. 30)

- SCONE Lab

  * Social Computing and Network Analysis

- 학위 논문

  * Improving Message-Passing and Representation Learning of Graph Neural Networks

Postdoc Researcher

- 한국에너지공과대학교 (2023. 09. 01 – 2023. 11.10)

  * 머신러닝 기반 효율적인 에너지 사용 관련 연구

- Arizona State University (2023. 11. 11 – current)

  * 머신러닝 기반 사막의 물 부족 현상 해결을 위한 연구

Projects

Recommender System

1. Knowledge transfer with domain disentanglement
2. LLM-based personalized recommendation
3. Hyperbolic cross-domain recommendation

Graph Theory

1. Mechanism of signed propagation
2. Feature sparsity in graph neural networks
3. Graph heterophily and edge weight retrieval

Projects – Recommender System

Title

Review-Based Domain Disentanglement without Duplicate Users or Contexts for Cross-Domain Recommendation, CIKM ’22

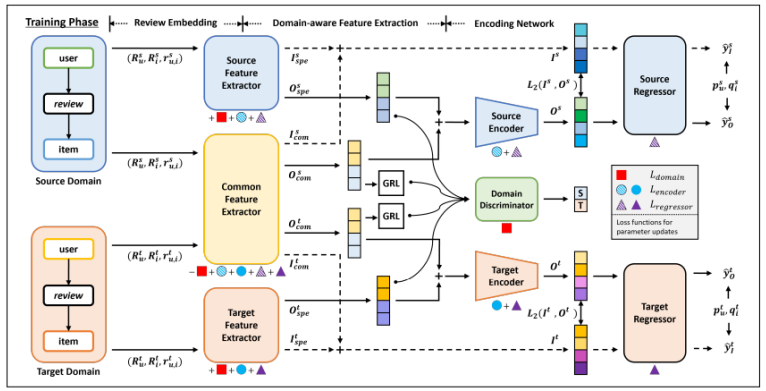
Objective

- 기존 지식 전이 기법들은 두 도메인에서 중복되는 정보를 기반으로 함 (동일한 사용자, 태그 등)

  * 이는 도메인이 다를 경우 지식 전이가 불가능함

- 본 논문은 사용자가 아이템을 구매하고 남긴 리뷰를 기반으로 정보를 전달

  * 이를 위해 기존 entropy-based domain disentanglement를 개선하여 성능 향상을 보임



Propsed Model

Target domain / Source domain	Office Product Baby	Office Product KS	Office Product TG	Office Product Yelp	Instant Video Baby	Instant Video KS	Instant Video TG	Instant Video Yelp
PMF		1.085				1.129		
NeuMF		0.974				1.014		
DeepCoNN		0.902				0.949		
NARRE		0.863				0.914		
AHN		0.859				0.892		
DANN	0.966	0.939	0.943	1.118	0.986	0.946	0.987	1.147
DAREC	0.989	0.988	0.972	0.994	1.060	1.045	1.043	1.073
DDTCDR	0.954	0.947	0.926	0.965	0.974	0.981	0.967	0.988
RC-DFM	0.834	0.839	0.828	0.841	0.878	0.855	0.868	0.872
CATN	0.875	0.872	0.873	0.876	0.915	0.906	0.892	0.919
MMT	0.815	0.820	0.822	0.856	0.862	0.855	0.878	0.871
SER	**0.789**	**0.815**	**0.810**	**0.806**	**0.852**	**0.833**	**0.855**	**0.847**

Performance (SER)

Projects – Recommender System

Title

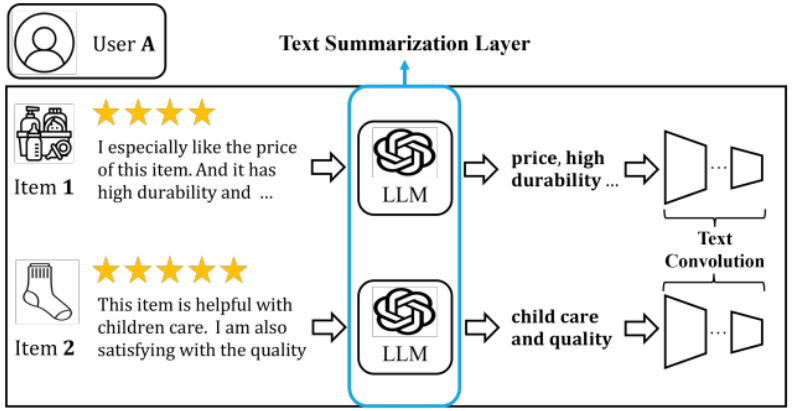
Improving the Text Convolution Mechanism with Large Language Model for Review-Based Recommendation, arXiv ’24

Objective

- 기존 추천시스템에서는 review data 분석을 위해 text convolution network (2D CNN)을 사용함

  * 이는 중요한 단어를 구분하기 어려운 문제점이 있음

- 본 논문에서는 이를 해결하기 위해 LLM 기반의 text summarization 이후 text convolution network를 적용하도록 개선하였으며, 아래 그림과 같이 information theory 관점에서 추가적인 분석을 진행함



Propsed Model

Before delving into the information gain, we introduce Shannon’s entropy [29] for the d-sized vector sets below:

$$H(T) = -\sum_{i=1}^{d} T_i log_d T_i. \quad (1)$$

Using the above equation, the information gain $IG(T, a)$ under separation condition (attribute) a is given by:

$$IG(T, a) = H(T) - H(T|a). \quad (2)$$

Information Theory

Projects – Recommender System

Title

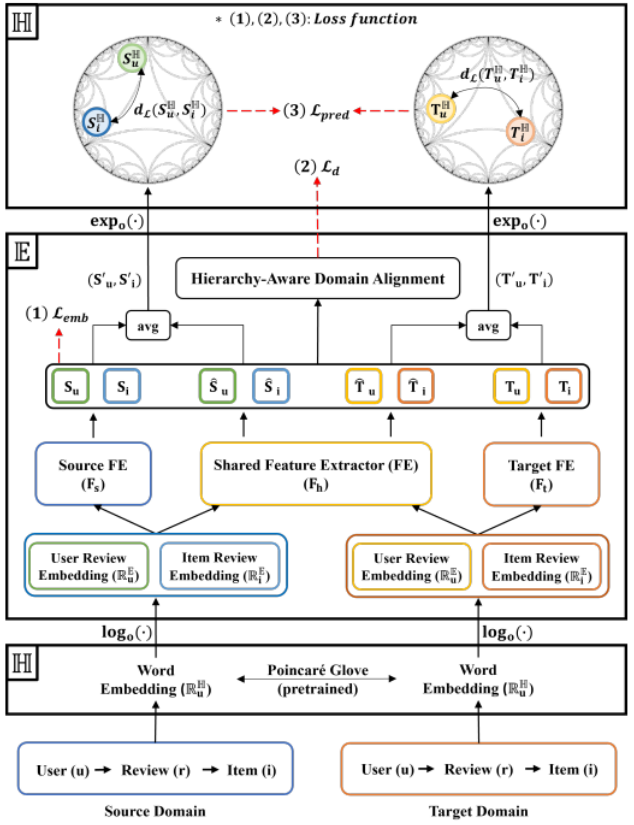
Review-Based Cross-Domain Recommendation via Hyperbolic Embedding and Hierarchy-Aware Domain Disentanglement, arXiv ’24

Objective

- Hyperbolic space에 embedding을 진행하는 최근 연구들은 single domain 위주로 진행되었음

- 본 논문에서는 hyperbolic space에서 지식 전이를 할 때, 기존 방법이 가지는 문제점을 지적하고 이를 해결하고자 함

* 구체적으로 personalized recommendation 및 지식 전이를 고려한 세 가지 loss function을 제안함



Propsed Model

Projects – Graph Theory

Title

Improving Signed Propagation of Graph Neural Network Under Multiple Classes, arXiv ’24

Objective

- Signed propagation을 multi-class dataset에서 사용할 경우 발생하는 문제점에 대한 새로운 이론적인 배경을 제공함

- 이를 기반으로 두 가지 calibration algorithm을 제안하였으며, heterophily graph에서 잘 동작함을 실험적으로 입증함

$$
\begin{aligned}
&\mathbb{E}\left[\cos(k'_{edge}, k) | k'_{edge}|\right] \\
&= \frac{|k|}{d}\sqrt{\left\{\sum_{j=0}^{c} -\cos(jc')^2\right\}^2 + \left\{\sum_{j=0}^{c} -\cos(jc')\sin(jc')\right\}^2} \\
&= \frac{|k|}{d}\sqrt{\left\{\sum_{j=0}^{c} \frac{-1-\cos(2\pi jc')}{2}\right\}^2 + \left\{\sum_{j=0}^{c} -\frac{\sin(2\pi jc')}{2}\right\}^2} \\
&= \frac{|k|}{d}\sqrt{\frac{c}{4} + \sum_{j=0}^{c}\cos(jc')^2} > \mathbb{E}\left[\cos(k', k)|k'|\right]
\end{aligned}
$$

New Theorem

Datasets	Actor	Chameleon	Squirrel
$\mathcal{H}_g$ (Eq. 1)	0.22	0.23	0.22
GCN	20.2 ± 0.4 (0.29)	49.3 ± 0.5 (0.19)	30.7 ± 0.7 (0.31)
GCN‡	21.7 ± 0.6 (0.62)	49.4 ± 0.6 (0.25)	31.5 ± 0.6 (0.58)
GAT	22.5 ± 0.3 (0.28)	47.9 ± 0.8 (0.17)	30.8 ± 0.9 (0.27)
GAT‡	23.2 ± 1.8 (0.43)	49.2 ± 0.4 (0.16)	30.3 ± 0.8 (0.40)
ACM-GCN	24.9 ± 2.0 (0.46)	49.5 ± 0.7 (0.20)	31.6 ± 0.4 (0.54)
HOG-GCN	21.5 ± 0.5 (0.37)	47.7 ± 0.5 (0.32)	30.1 ± 0.4 (0.51)
JacobiConv	25.7 ± 1.2 (0.30)	**52.8** ± 0.9 (0.21)	32.0 ± 0.6 (0.37)
GloGNN	26.6 ± 0.7 (0.38)	48.2 ± 0.3 (0.27)	28.8 ± 0.8 (0.43)
PTDNet	21.5 ± 0.6 (0.33)	50.6 ± 0.9 (0.17)	32.1 ± 0.7 (0.34)
PTDNet‡	22.7 ± 0.6 (0.19)	50.9 ± 0.3 (0.15)	**32.3** ± 0.5 (0.30)
GPRGNN	25.4 ± 0.5 (0.55)	49.1 ± 0.7 (0.25)	30.5 ± 0.6 (0.36)
GPRGNN‡	**28.1** ± 1.3 (0.33)	51.0 ± 0.4 (0.18)	31.8 ± 0.4 (0.16)
FAGCN	25.2 ± 0.8 (0.66)	46.5 ± 1.1 (0.25)	30.4 ± 0.4 (0.64)
FAGCN‡	27.6 ± 0.5 (0.42)	48.8 ± 0.7 (0.13)	31.3 ± 0.5 (0.37)
GGCN	22.5 ± 0.5 (0.47)	48.5 ± 0.7 (0.15)	30.2 ± 0.7 (0.40)
GGCN‡	24.6 ± 0.4 (0.26)	50.0 ± 0.4 (0.07)	31.1 ± 0.6 (0.15)

Performance Analysis

Projects – Graph Theory

Title

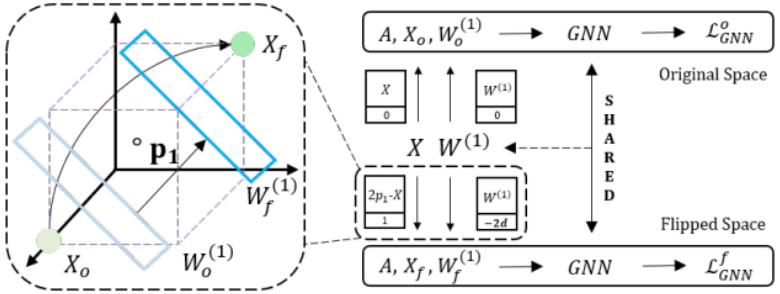
Perturb Initial Features: Generalization of Neural Networks Under Sparse Features for Semi-supervised Node Classification, arXiv ’23

Objective

- 기존 message-passing scheme 개선에만 집중하던 방법들과 달리, node’s initial feature에 존재하는 zero element 값에 대한 문제점을 지적함

- Zero element가 대부분일 경우, gradient vanishing 현상 및 first hyperplane overfitting 현상으로 parameter가 제대로 학습되지 않음

- 본 논문에서는 간단한 flipping strategy를 기반으로 30% 이상의 성능 향상이 가능함을 보임



Proposed Method

Datasets	Cora	Citeseer	Pubmed	Actor	Chameleon	Squirrel	Cornell	Texas	Wisconsin
z (Eq. 4)	0.59	0.41	0.96	0.21	1.0	1.0	0.62	0.41	0.53
Homophily (Eq. 25)	0.81	0.74	0.8	0.22	0.23	0.22	0.11	0.06	0.16
MLP	53.2 ±0.5%	53.7 ±1.7%	69.7 ±0.4%	27.9 ±1.1%	41.2 ±1.8%	26.5 ±0.6%	60.1 ±1.2%	65.8 ±5.0%	73.5 ±5.4%
GCN†	79.1 ±0.7%	67.5 ±0.3%	77.8 ±0.2%	20.4 ±0.6%	49.4 ±0.7%	31.8 ±0.9%	39.4 ±4.3%	47.6 ±0.7%	40.5 ±1.9%
Ortho-GCN†	80.6 ±0.4%	69.5 ±0.3%	76.9 ±0.3%	21.4 ±1.6%	46.7 ±0.5%	31.3 ±0.6%	45.4 ±4.7%	53.1 ±3.9%	46.6 ±5.8%
GAT	80.1 ±0.6%	68.0 ±0.7%	78.0 ±0.4%	22.5 ±0.3%	46.9 ±0.8%	30.8 ±0.9%	42.1 ±3.1%	49.2 ±4.4%	45.8 ±5.3%
GATv2	79.5 ±0.5%	67.4 ±0.6%	76.2 ±0.5%	22.1 ±2.0%	48.3 ±0.4%	28.9 ±1.2%	38.0 ±3.8%	52.5 ±1.7%	41.7 ±5.1%
GIN	77.3 ±0.8%	66.1 ±0.6%	77.1 ±0.7%	24.6 ±0.8%	49.1 ±0.7%	28.4 ±2.2%	42.9 ±4.6%	53.5 ±3.0%	38.2 ±1.5%
APPNP	81.2 ±0.4%	68.9 ±0.3%	79.0 ±0.4%	21.5 ±0.2%	45.0 ±0.5%	30.3 ±0.6%	49.8 ±3.6%	56.1 ±0.2%	45.7 ±1.7%
GCNII	80.8 ±0.7%	69.0 ±1.4%	78.8 ±0.4%	26.1 ±1.2%	45.1 ±0.5%	28.1 ±0.7%	62.5 ±0.5%	69.3 ±2.1%	63.2 ±3.0%
H_2GCN	79.5 ±0.6%	67.4 ±0.5%	78.7 ±0.3%	25.8 ±1.2%	47.3 ±0.9%	31.1 ±0.5%	59.8 ±3.7%	66.3 ±4.6%	61.5 ±4.4%
P-reg†	80.0 ±0.8%	69.2 ±0.7%	77.4 ±0.4%	20.9 ±0.5%	49.1 ±0.1%	**33.6*** ±0.4%	44.9 ±3.1%	58.5 ±4.2%	53.7 ±2.6%
FAGCN	81.0 ±0.3%	68.3 ±0.6%	78.9 ±0.4%	26.7 ±0.8%	46.8 ±0.6%	29.9 ±0.5%	46.5 ±1.7%	53.8 ±1.2%	51.0 ±4.1%
GPR-GNN	82.2 ±0.4%	70.1 ±0.8%	79.3 ±0.3%	25.1 ±0.5%	50.9 ±0.8%	30.4 ±0.4%	53.1 ±1.6%	61.2 ±0.9%	62.4 ±1.2%
ACM-GCN	80.2 ±0.8%	68.3 ±1.1%	78.1 ±0.5%	24.9 ±2.0%	49.5 ±0.7%	31.6 ±0.4%	55.6 ±3.3%	58.9 ±2.6%	61.3 ±0.5%
JacobiConv	81.9 ±0.6%	69.6 ±0.8%	78.5 ±0.4%	25.7 ±1.2%	**52.8*** ±0.9%	32.0 ±0.6%	55.3 ±3.4%	57.7 ±3.6%	53.4 ±1.6%
GloGNN	82.4 ±0.3%	70.3 ±0.5%	79.3 ±0.2%	26.6 ±0.7%	48.2 ±0.3%	28.8 ±0.8%	56.8 ±1.1%	63.0 ±2.9%	59.5 ±1.3%
Flip-MLP	61.4 ±0.7%	60.3 ±0.5%	74.1 ±0.5%	**35.9*** ±0.5%	43.5 ±1.2%	28.5 ±0.8%	**70.5*** ±4.1%	**79.2*** ±3.6%	**80.5*** ±5.1%
vs MLP (+ %)	+ 15.4 %	+ 12.1 %	+ 6.3 %	**+ 28.7** %	+ 2.3 %	+ 2.1 %	**+ 17.3** %	**+ 20.4** %	**+ 9.5** %
α, β	1, 0.1	1, 1	1, 0.01	0.1, 0.1	1, 1	1, 1	0.1, 0.1	0.1, 0.01	1, 1
Flip-GCN†	82.7 ±0.5%	72.4 ±0.4%	79.2 ±0.2%	28.6 ±0.3%	50.4 ±0.5%	32.4 ±0.3%	49.4 ±0.6%	62.2 ±1.8%	52.3 ±2.4%
vs GCN (+ %)	+ 4.6 %	+ 7.3 %	+ 1.8 %	+ 40.2 %	+ 2.0 %	+ 1.9 %	+ 20.4 %	+ 30.7 %	+ 29.1 %
α, β	0.1, 0.01	0.01, $1e^{-3}$	1, 0.01	$1e^{-3}$, $1e^{-3}$	1, $1e^{-3}$	1, 0.01	1, $1e^{-3}$	1, $1e^{-3}$	0.01, $1e^{-3}$
Flip-GAT	83.1 ±0.6%	72.8 ±0.4%	78.5 ±0.2%	30.3 ±0.8%	48.3 ±0.3%	31.9 ±0.5%	51.9 ±1.4%	61.0 ±1.1%	54.5 ±2.1%
vs GAT (+ %)	+ 3.7 %	+ 7.1 %	+ 0.6 %	+ 34.7 %	+ 3.0%	+ 3.6 %	+ 20.2 %	+ 22.0 %	+ 19.0 %
α, β	1, 0.1	0.01, $1e^{-3}$	1, $1e^{-3}$	0.1, 0.1	1, 0.1	1, 1	0.1, $1e^{-3}$	0.1, $1e^{-3}$	0.1, $1e^{-3}$
Flip-GIN	79.5 ±0.8%	69.3 ±0.9%	78.3 ±0.5%	28.5 ±0.5%	50.0 ±0.7%	28.6 ±1.1%	48.1 ±2.3%	57.3 ±1.9%	53.9 ±1.7%
vs GIN (+ %)	+ 2.8 %	+ 4.8 %	+ 1.6 %	+ 15.9 %	+ 1.8 %	+ 1 %	+ 12.1 %	+ 7.1 %	+ 39.3 %
α, β	$1e^{-3}$, $1e^{-3}$	0.1, $1e^{-3}$	1, 0.1	0.01, 0.01	1, $1e^{-3}$	1, 0.1	1, $1e^{-3}$	1, $1e^{-3}$	0.01, $1e^{-3}$
Flip-APPNP	**83.3*** ±0.3%	**72.9*** ±2.3%	**79.6*** ±0.4%	31.5 ±0.5%	45.6 ±0.5%	30.4 ±0.4%	56.8 ±1.5%	62.8 ±0.5%	61.0 ±2.8%
vs APPNP (+ %)	**+ 2.6** %	**+ 5.8** %	**+ 0.7** %	+ 46.5 %	+ 1.3 %	+ 0.1 %	+ 14.1 %	+ 11.9 %	+ 33.5 %
α, β	1, 0.01	0.1, $1e^{-3}$	1, 0.01	0.01, 0.01	1, 0.01	0.1, 0.01	1, 0.01	1, $1e^{-3}$	0.1, 0.01

Performance Analysis

Projects – Graph Theory

Title

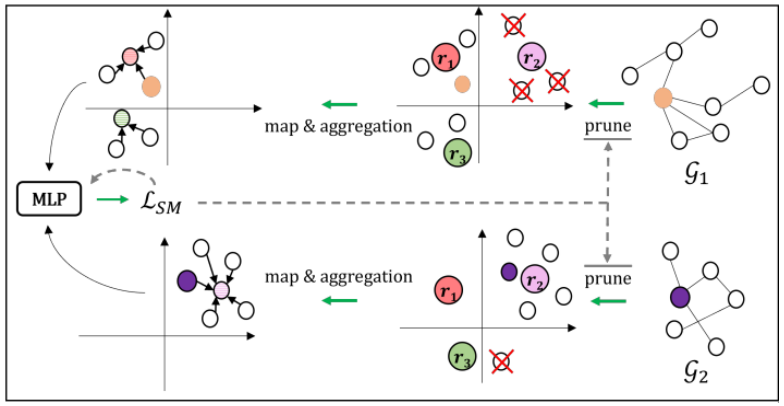
Finding Heterophilic Neighbors via Confidence-based Subgraph Matching for Semi-supervised Node Classification, CIKM '22

Objective

- Edge weight를 구하는 것은 message-passing에 있어 매우 중요한 과정임

- 기존 방법들은 단순히 두 노드의 feature similarity를 측정하여 이를 해결

- 본 논문에서는 해당 방법이 불확실함을 지적한 뒤, subgraph 기반의 보다 정확한 edge weight retrieval 방법을 제안함



Proposed Method

Datasets	Actor	Chameleon	Squirrel
Hom. ratio (h)	0.22	0.23	0.22
MLP	**27.9*** ±1.1%	41.2 ±1.8%	26.5 ±0.6%
GCN	21.4 ±0.6%	49.4 ±0.7%	33.1 ±1.2%
DropEdge†	21.9 ±0.4%	49.2 ±0.8%	31.0 ±1.4%
GAT	23.2 ±0.8%	48.2 ±0.6%	30.1 ±0.9%
GIN	23.6 ±0.5%	40.1 ±3.7%	23.0 ±2.4%
APPNP	21.7 ±0.2%	45.2 ±0.7%	30.6 ±0.7%
GCNII	25.7 ±0.4%	45.1 ±0.5%	28.8 ±0.3%
GAM†	21.7 ±0.3%	49.5 ±0.5%	33.9 ±0.4%
H_2GCN	22.8 ±0.3%	46.6 ±1.2%	29.8 ±0.7%
FAGCN	26.9 ±0.4%	46.7 ±0.6%	29.5 ±0.7%
PTDNet†	21.5 ±0.5%	49.7 ±0.9%	32.6 ±0.7%
Ours†	27.7 ±0.4%	**52.3*** ±0.4%	**35.7*** ±0.5%

Performance Analysis

Other Projects (이력서 2페이지를 참고해 주시면 감사하겠습니다)

Selection Criteria and Assigned Weightage for Identifying Potential Locations Wetland

- 사막 기후에서 핵심인 wetland (습지)를 찾기 위하여 knowledge transfer 기법을 제안

Causal Discovery of Agricultural Mgmt. and Reservoir Op. Induced Water Quality Change

- Causal-inference 기반의 수질 관리 및 물의 이동 방향 예측

Tracking footprints with graph neural networks for the reduction of virus spread

- 코로나 바이러스의 확산 방지를 위해 각 유저의 행동 패턴 분석

Personalized recommendation based on the user's purchasing histories and social network

- 유저의 구매 기록 및 사회 관계망을 기반으로 개인화된 추천 방법 제안

Future Work

LLM-based Recommender System

- ChatGPT, LLaMa 등 언어 모델이 발전함에 따라 대규모 데이터에서 추출한 사용자 특성의 정확도가 크게 증가함

- 특히 time-series analysis에 강점을 보이는 LLM을 user-item interaction에 접목하여 personalized recommendation quality를 개선하는 연구를 진행 중

Graph Theory

- Information propagation이 항상 성능 향상을 가져온다는 기존 통념과 달리, 특정 상황에서는 propagation을 막는 것이 유리함을 증명

- Sub-stochastic matrix의 spectral radius 분석 및 disconnected component의 smoothing effect를 기반으로 새로운 방법을 제안하려 함

Summary

1. Personalized ranking, click-through rate prediction 등의 연구 수행 경험이 다수

2. PyTorch 기반의 End-to-end training 및 전체 네트워크의 sub-module 개발 경험 다수

3. 한정된 GPU에서 대규모 데이터 실험을 위해 resource optimization을 진행한 경험

4. 오픈소스가 아닌, 논문 작성을 위해 새로운 방법을 제시하였으며, 직접 모델을 설계하고 구축한 경험 다수

Thank you